

MIART · MICCAI 2026

DoMa-Seg

Dose Map Guidance for Segmentation in Head & Neck Radiotherapy

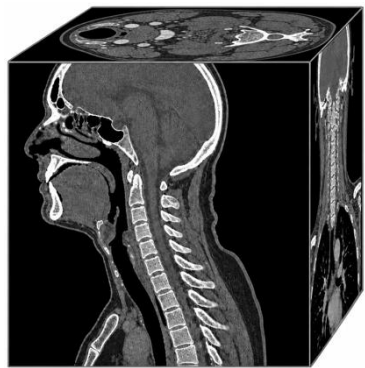
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01 — Background

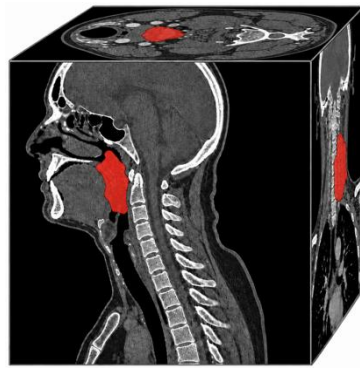
Segmentation & Dose Planning in Radiotherapy

- Radiotherapy planning workflow



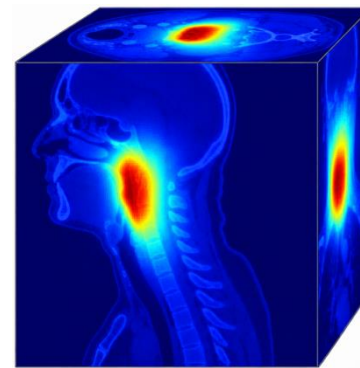
CT

Segment



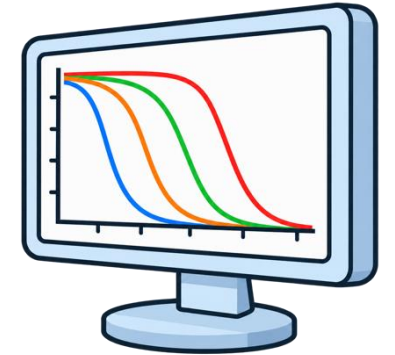
Contours

Dose
planning



Dose distribution

Evaluate

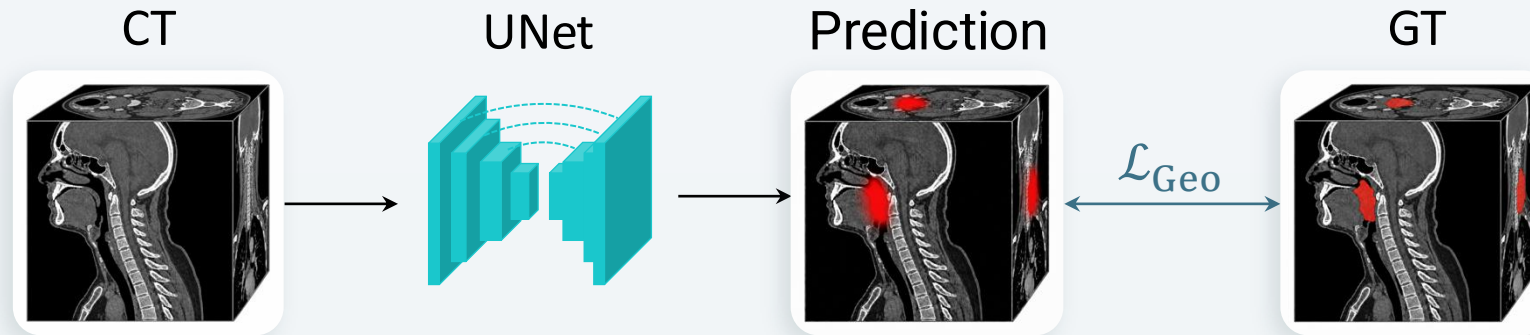


DVH endpoints

01 — Background

Current Auto-Segmentation Paradigm

- **Geometry-based Segmentation (GeoSeg)**
 - Trained on Geometry-based losses, e.g., Dice and cross-entropy

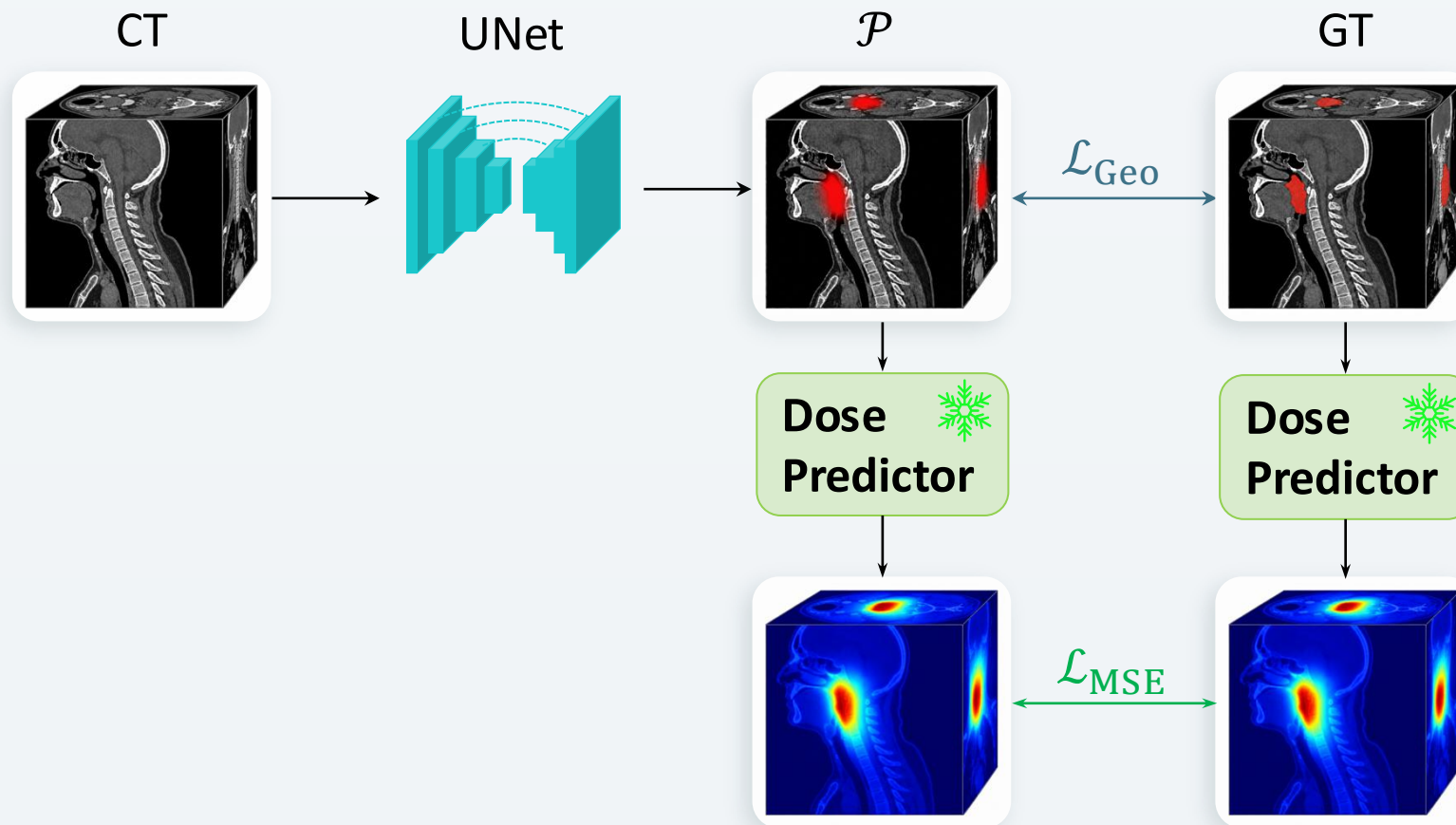


The issue: geometric overlap does not guarantee dosimetric acceptance.

01 — Background

Current Auto-Segmentation Paradigm

- **Dose Predictor based Segmentation (DoPre-Seg)**
 - Integrate dose loss using a pre-trained dose predictor (MICCAI 2023)



The issues:

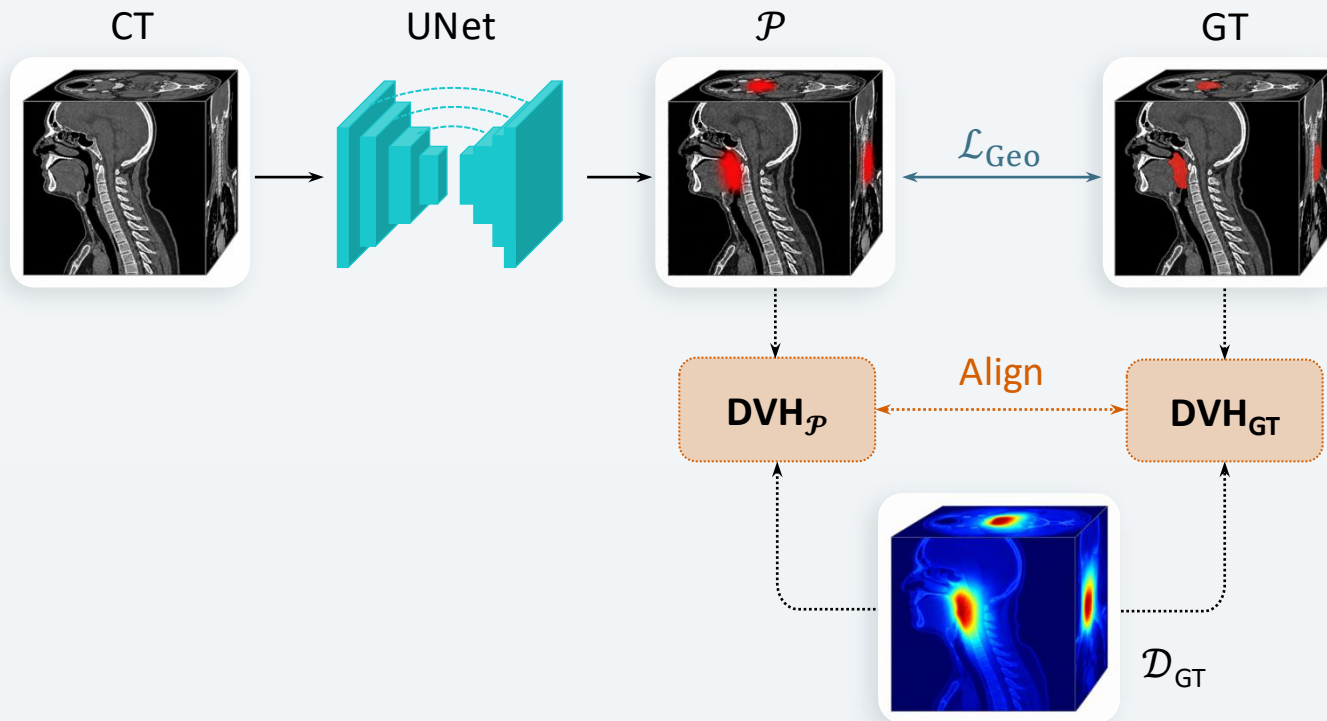
- Predictor-dependent
- Computation overhead
- Memory overhead
- Cascaded errors

02 — Method

Dose Map Guided Segmentation (DoMa-Seg)

- Key idea

- Use the retrospective dose maps as supervision to integrate dose awareness



Advantages

- No need for extra network
- Computational efficiency
- Avoid cascaded errors

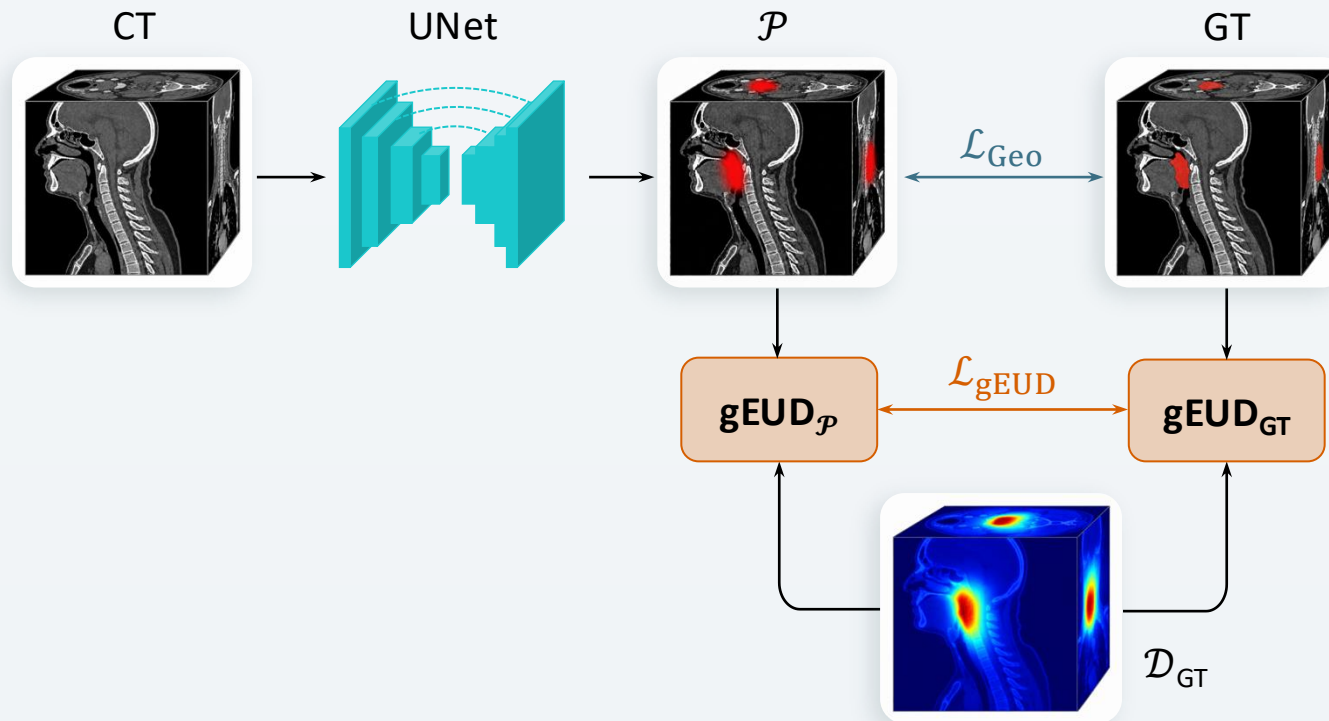
The issue: DVH parameter calculation is non-differentiable

02 — Method

Dose Map Guided Segmentation (DoMa-Seg)

- Key idea

- Use the retrospective dose maps as supervision to integrate dose awareness



Radiobiological concept

Generalized Equivalent Uniform Dose (gEUD) to approximate DVH parameters.

Differentiable gEUD surrogate

Make it differentiable w.r.t. predicted probability maps for training.

Refs: gEUD: Niemierko, Med. Phys. 1997 & 1999.

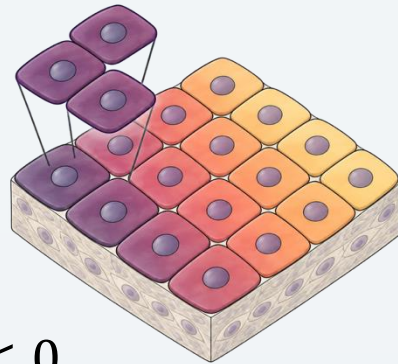
02 — Method

Dose Map Guided Segmentation (DoMa-Seg)

- Differentiable gEUD loss

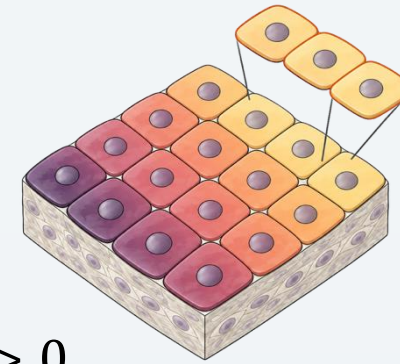
$$\text{gEUD}(S) = \left[\frac{\sum_j S_j \mathcal{D}_j^\gamma}{\sum_j S_j} \right]^{\frac{1}{\gamma}}$$

$S = \mathcal{P}$ or $\mathcal{Y}$; $\mathcal{D}$ = clinical dose; γ : scheduling hyperparameter



$\gamma < 0$

- Emphasize low-dose regions



$\gamma > 0$

- Emphasize high-dose regions

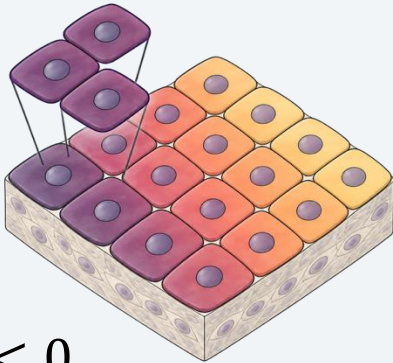
02 — Method

Dose Map Guided Segmentation (DoMa-Seg)

- Differentiable gEUD loss

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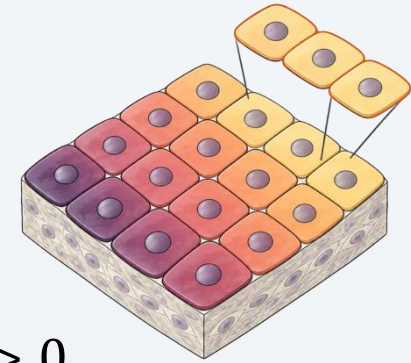
$S = \mathcal{P}$ or $\mathcal{Y}$; $\mathcal{D}$ = clinical dose; γ : scheduling hyperparameter



$\gamma < 0$

- Emphasize low-dose regions

$$\mathcal{L}_{\text{gEUD}} = |\text{gEUD}_{\text{GT}} - \text{gEUD}_{\text{Pred}}|$$



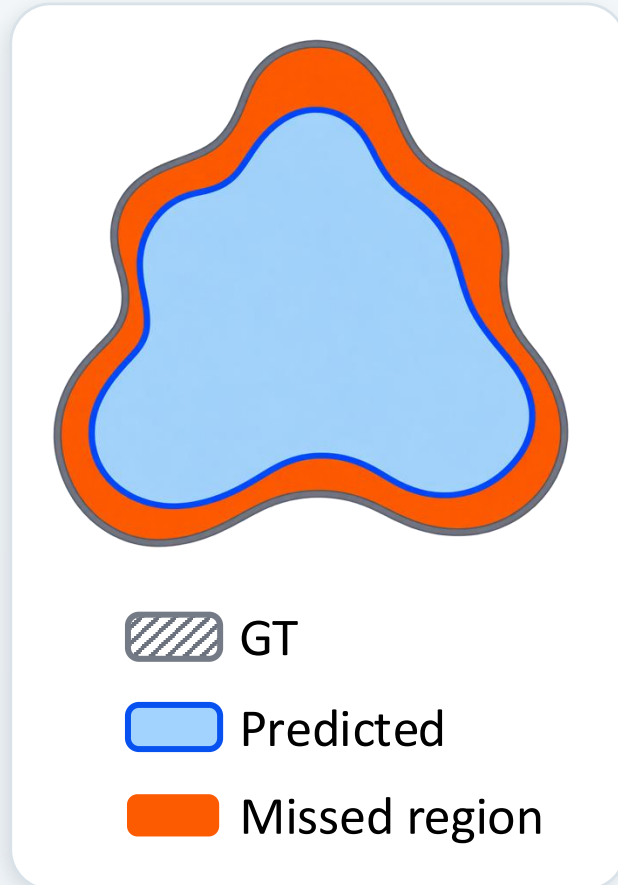
$\gamma > 0$

- Emphasize high-dose regions

02 — Method

Dose Map Guided Segmentation (DoMa-Seg)

- Preserve target coverage by Tversky loss



$$\mathcal{L} = \underbrace{\mathcal{L}_{gEUD}}_{\text{Dose confinement}} + \underbrace{\mathcal{L}_{Dice}}_{\text{Geometric agreement}} + \underbrace{\mathcal{L}_{Focal}}_{\text{Target coverage}} + \mathcal{L}_{Tversky}$$

$$\mathcal{L}_{Tversky} = 1 - \frac{TP}{TP + \alpha \cdot FP + \beta \cdot FN} \quad \text{PTV: } \beta > \alpha$$

Greater weight on false negatives (missed target voxels)

03 — Experimental Setup

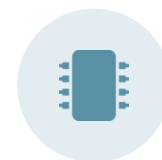
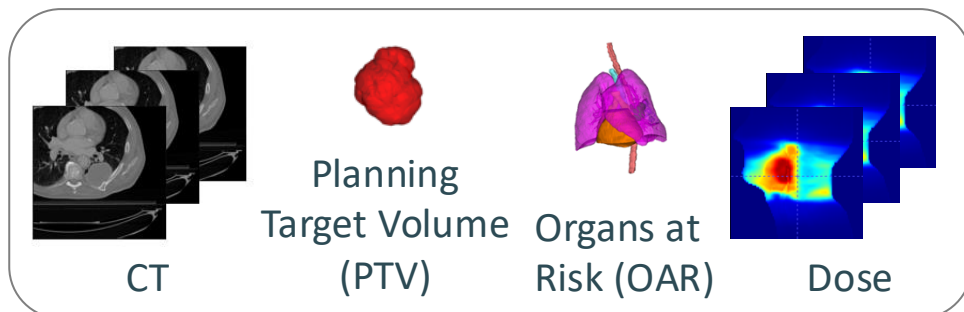
Data & Evaluation



Dataset

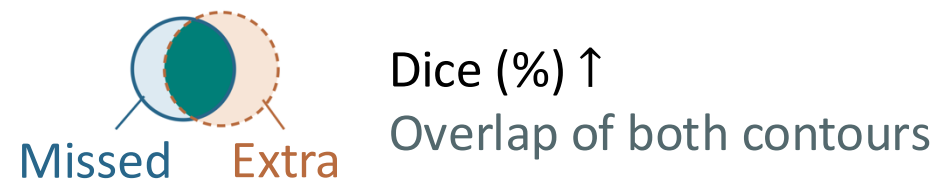
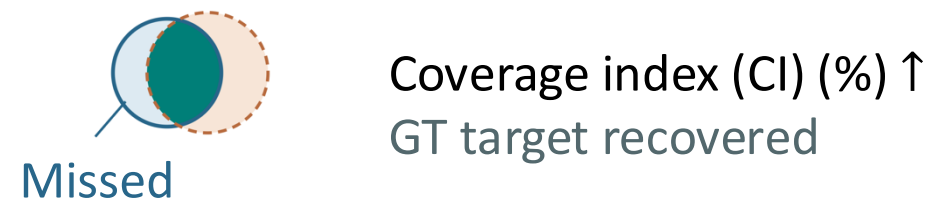
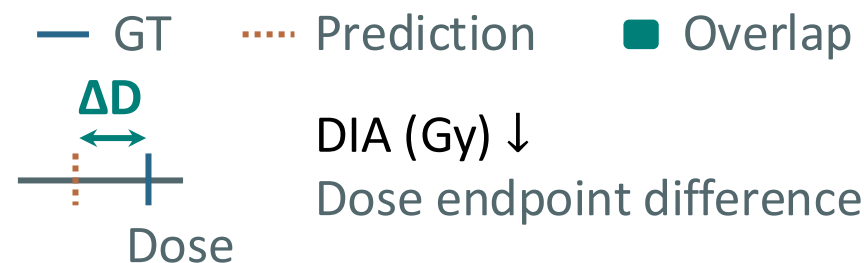
GDP-HMM Head & Neck Challenge

- 400 train / 70 test, across 5 centers
- 11 structures: 1 PTV + 10 OARs



Evaluation Metrics

Dosimetric evaluation

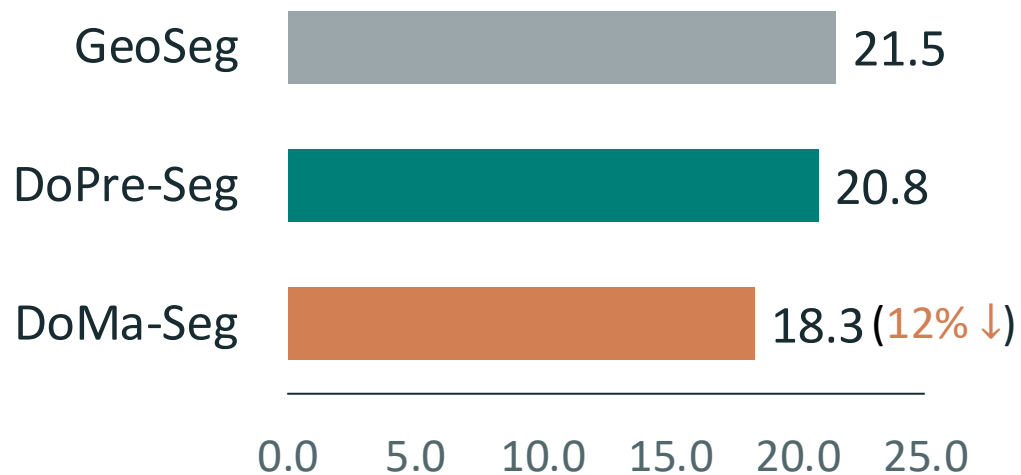


Refs: Dataset: GDP-HMM (Gao et al. 2025). ABC: Poel et al. 2025.

04 — Results

Lower Target Dose Discrepancy with Comparable Overlap

PTV DIA_{99%} (Gy) ↓



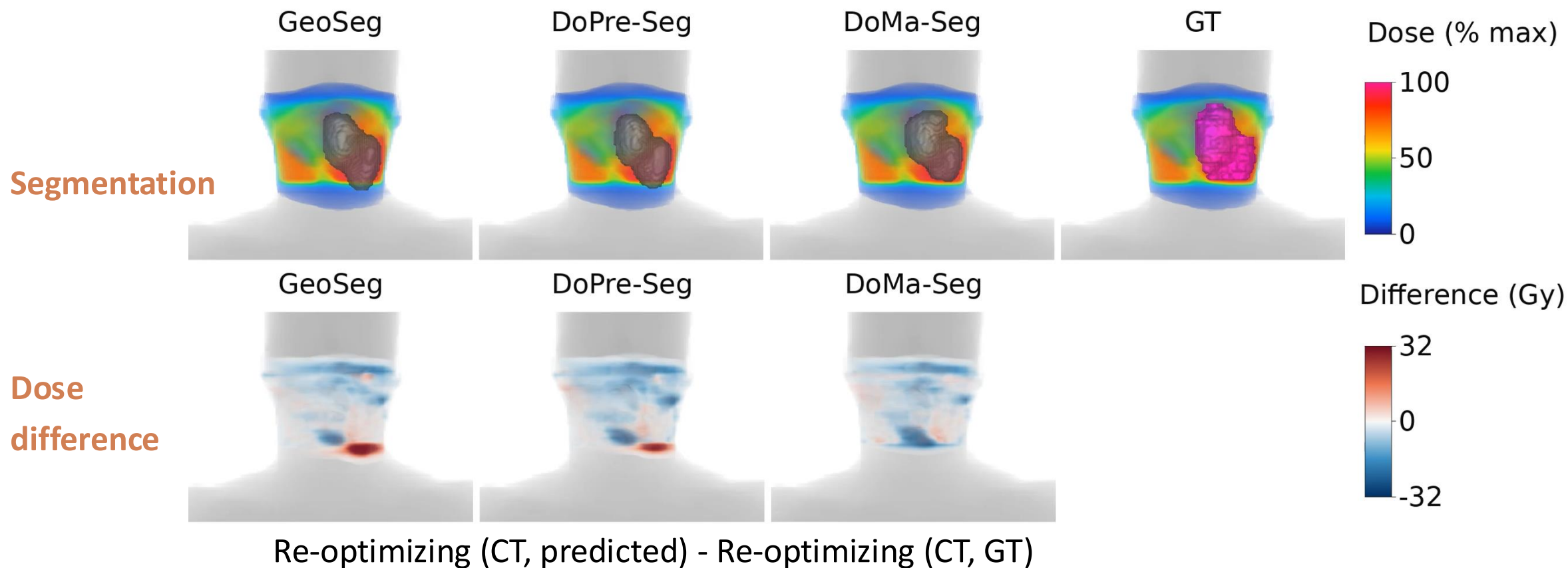
Overlap and coverage

	CI (%)	Dice (%)
GeoSeg	61.9	62.7
DoPre-Seg	61.7	62.8
DoMa-Seg	64.1	63.4

+2.4 pp CI +0.6 pp Dice

04 — Results

Volumetric Segmentation and Re-optimized Dose Difference



05 — Conclusion

Closing the Geometric–Dosimetric Gap

- ✓ Dose-aware training via a differentiable gEUD loss — no extra network.
- ✓ 12% lower $\text{DIA}_{99\%}$ (dose discrepancy) than baseline methods.
- ✓ A potentially practical step toward dosimetrically reliable radiotherapy auto-contouring.
- ✓ Clinical evaluation of the proposed method is ongoing.

Thank you for listening!

Questions & discussion

Jihe Li · University of Bern